



**FACULTY OF COMPUTING AND INFORMATION SYSTEMS
CSDS 336 - FUNDAMENTALS OF MACHINE LEARNING**

BRAIN TUMOR DETECTION MODEL ANALYSIS REPORT

**REPORTED BY
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LEVEL 300**

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1 INTRODUCTION

Brain tumors are abnormal growths of cells in the brain which can be life-threatening if not diagnosed early. The World Health Organization claims that around 400,000 individuals around the world are diagnosed with a brain tumor (Abdalla et al., 2018). In the US, it has been estimated that around 86,970 new cases of brain tumors will be diagnosed in 2019 (Abdalla et al., 2018). Automated detection using medical imaging can assist radiologists in faster diagnosis. This project explores using convolutional neural networks (CNNs) to classify brain MRI images into pituitary tumor and no-tumor categories.

Although the initial studies on the use of machine learning in predicting brain tumors show the potential of this technology, they also highlight the limitations of this approach like efficiency of model and model accuracy ((Hossain et al., 2019), (Brindha et al., 2021)). By conducting a follow-up study, we can hopefully address these issues and contribute to the advancement of this technology. The implementation of machine learning in brain tumor treatment could represent a massive breakthrough in patient outcome prediction and could potentially help in saving lives (Sharma et al., 2022).

1.1 BACKGROUND OF STUDY

Brain MRI images are commonly used in diagnosing brain tumors. Traditional diagnosis requires manual inspection by radiologists, which, based on precedence, is time-consuming and subject to human error (Sharma et al., 2022). Machine learning, especially deep learning with CNNs, has shown potential in accurately classifying medical images.

This new found technology, along with its methodologies and problem-solving algorithms, has proven to boost efficiency and accuracy of early detection of brain tumors.

1.2 PROBLEM STATEMENT

The problem identified here is how manual inspection of MRI scans can be time-intensive and prone to errors. Early detection of tumor is very crucial to saving the lives of the infected patients. Currently, most methods rely on CATs and MRI to find brain cancers, which have limitations in terms of accuracy (Sharma et al., 2022). Furthermore, these tools do not provide enough precision as needed isn't critical (Brindha, P and other writers., 2021).

Also, they are very likely to bring subjectivity into their analytical process, hence highlighting the need for more precise and neutral methods (Sharma et al., 2022). The challenge is to build a reliable machine learning model that can automatically classify brain MRI images with high accuracy and generalization.

1.3 AIM

The aim is to develop and compare multiple CNN models from scratch for classifying brain MRI images into pituitary tumor and no-tumor classes. This project is focused on developing a method that uses deep learning and MRI data. This involves training neural networks on small sets of data, such as images of brain tumors, in order to study what the best model is for such dataset.

1.4 OBJECTIVES

The objectives of this project are;

1. Build three CNN models from scratch and compare their performance.
2. Evaluate models using AUC, precision, recall and accuracy. As well as a validation curve.

1.5 SIGNIFICANCE OF PROJECT

This project demonstrates the use of deep learning in medical image classification by

highlighting the effectiveness of CNNs in assisting radiologists. It also provides insight into the trade-offs between model complexity, regularization, and dataset size, which is useful for future ML projects in healthcare.

1.6 PRELIMINARY LITERATURE REVIEW

A primary brain tumor is an abnormal growth that starts in the brain and usually does not spread to other parts of the body (University of Colorado, 2013). Primary brain tumors may be benign or malignant. Pituitary adenomas (PA) are benign tumors which account for being the second most common intracranial tumors after meningiomas (Ibrahim et al., 2022). In diagnosing for brain tumors, the doctor will obtain your personal and family medical history and perform a complete physical examination.

In addition to checking your general health, the doctor performs a neurological exam to check mental status and memory, cranial nerve function (sight, hearing, smell, tongue and facial movement), muscle strength, coordination, reflexes, and response to pain (University of Colorado, 2013). In addition to these procedures, an MRI scan is done for a clearly identification of the tumor. If a diagnosis cannot be made clearly from the scans, a biopsy may be performed to determine what type of tumor is present. Biopsy is a procedure to remove a small amount of tumor to be examined by a pathologist under a microscope (University of Colorado, 2013).

For a wide range of healthcare research and applications, the convolutional neural network models had always demonstrated to acquire higher-grade results (Sharma et al., 2022). Still, building these pretrained convolutional neural network models from scratch had always been strenuous for prediction of this neurological disease due to restricted access of computed tomography (CT) image (Gu et al., 2021).

1.7 METHODOLOGY

1. Dataset Acquisition
2. Data Preprocessing
3. Model Selection and Development
4. Training and Hyperparameter Tuning
5. Evaluation
6. Comparative Analysis and Inference

Conceptual Framework

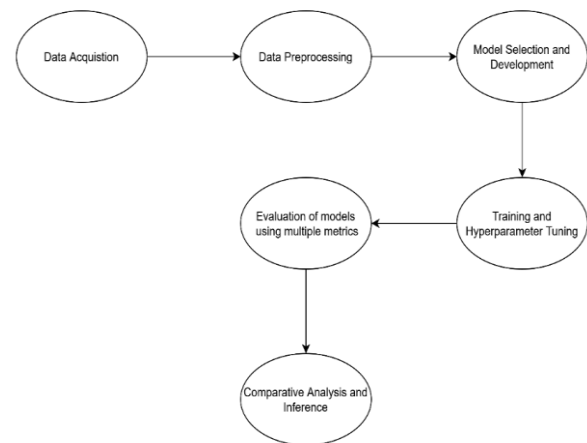


Figure 1. Diagram of Conceptual framework

1.7.1 DATASET ACQUISITION

The dataset used for this project was obtained from Kaggle, a public repository of pre-gathered and structured datasets.

Specifically, the Brain Tumor Classification MRI dataset was sourced from Kaggle at: <https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri/>

The dataset has four classes namely:

- glioma_tumor
- meningioma_tumor
- pituitary_tumor
- no_tumor

As indicated by their names, the classes are self-descriptive. However, in this experiment, only the **pituitary_tumor** and **no_tumor** classes were selected.

This decision was made due to the limited size of the dataset, which may not be

sufficient for effective generalization in a multi-class classification setting. By reducing the task to binary classification, the models are able to learn more discriminative features and achieve more reliable performance.

1.7.2 DATA PREPROCESSING

The preprocessing technique used here was image augmentation for increasing the size of the dataset for better generalization. This included:

- Rescaling
- Flipping
- Resizing
- Zooming
- Rotation

The intensity of image augmentation was done differently for each model to better comparative analysis.

1.7.3 MODEL SELECTION AND DEVELOPMENT

Three models were made in this project using the Convolutional Neural Networks (CNN) algorithm. The following table shows the models with their architectural layers:

Model	Architecture	Regularization / Augmentation
Model 1 (Base CNN)	Shallow CNN with single convolutional block	Rescaling only
Model 2 (Tuned CNN)	Moderately deep CNN with two convolutional blocks	Dropout + heavy augmentation
Model 3 (Deeper CNN)	Deeper CNN with three convolutional layers	Dropout + mild augmentation (resizing and rescaling only)

Table 1. Structure of all models

1.7.4 HYPERPARAMETER TUNING

Hyperparameters for Model 1

- Learning rate = 0.001
- Batch size = 8
- Epochs = 15
- Number of filters on 1st convolutional layer = 16

Hyperparameters for Model 2

- Learning rate = 0.0001
- Batch size = 8
- Epochs = 15
- Number of filters on 1st convolutional layer = 16
- Number of filters on 2nd convolutional layer = 32
- Dropout for 1st convolutional layer = 0.25
- Dropout for 2nd convolutional layer = 0.50

Hyperparameters for Model 3

- Learning rate = 0.0001
- Batch size = 8
- Epochs = 15
- Number of filters on 1st convolutional layer = 16
- Number of filters on 2nd convolutional layer = 32
- Number of filters on 3rd convolutional layer = 64
- Dropout for 2nd convolutional layer = 0.2
- Dropout for 3rd convolutional layer = 0.3

1.7.5 EVALUATION AND INFERENCE

Metric	Model 1	Model 2	Model 3
Training Accuracy	1.0000	0.9182	0.8813
Training AUC	1.0000	0.9098	0.9195
Training Precision	1.0000	0.9758	0.8723
Training Recall	1.0000	0.9647	0.9661

Table 2. Training Evaluation Metrics

Metric	Model 1	Model 2	Model 3
Validation Accuracy	0.9497	0.7318	0.6089
Validation AUC	0.9983	0.8246	0.6403
Validation Precision	1.0000	0.9333	0.5370
Validation Recall	0.8784	0.3784	0.3919

Table 2.1 Validation Evaluation Metrics

Validation Curves for Each Model

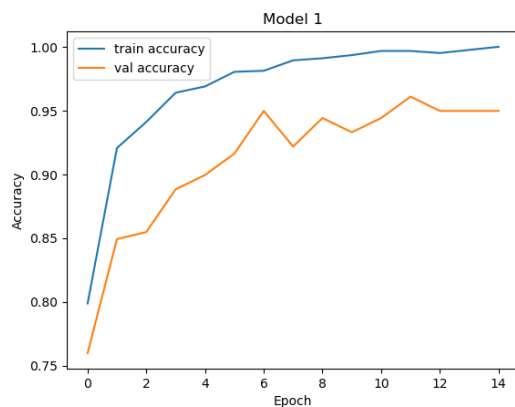


Figure 2. Model 1 Validation Curve

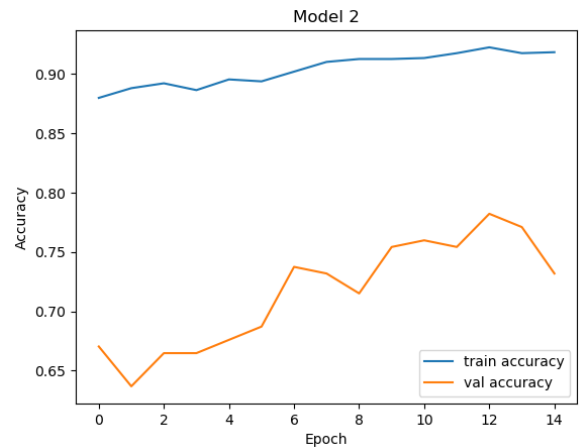


Figure 2.1 Model 2 Validation Curve

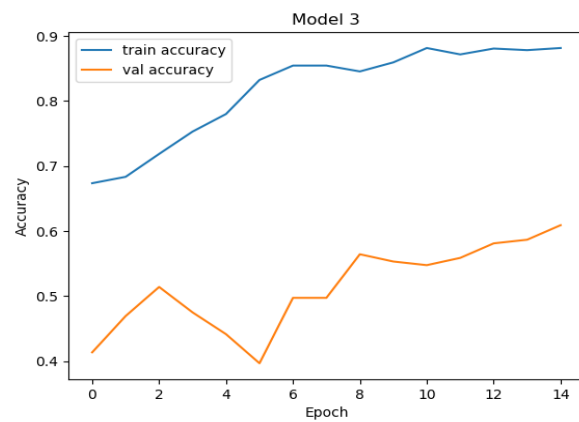


Figure 2.2 Model 3 Validation Curve

ROC Curve for Each Model

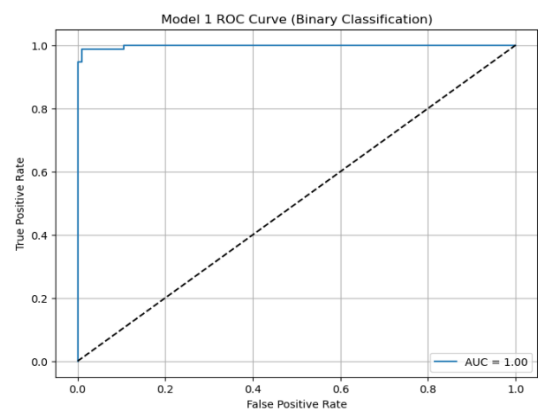


Figure 3. Model 1 ROC Curve

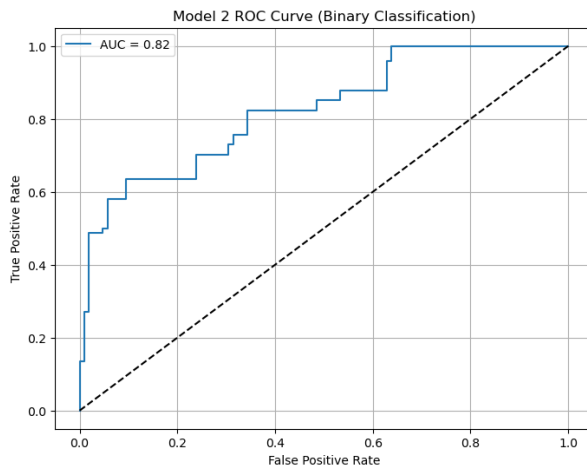


Figure 3.1 Model 2 ROC Curve

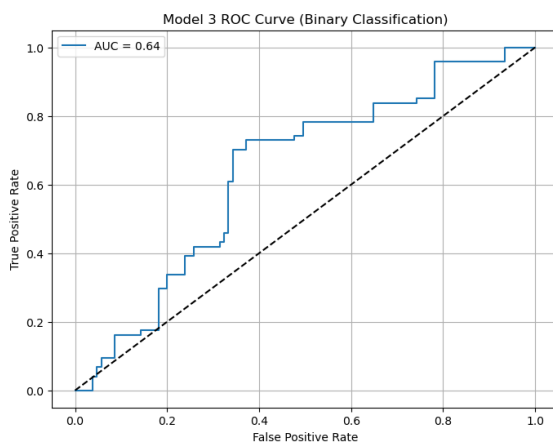


Figure 3.2 Model 3 ROC Curve

Comparative Analysis

The three models differ primarily in depth and preprocessing. Model 1 is a simple baseline CNN, Model 2 introduces an additional convolutional layer and heavy augmentation, while Model 3 is a deeper architecture with mild augmentation.

Model 1 achieved near-perfect training accuracy, while Model 2 and 3 showed slightly lower training accuracy due to increased complexity and the regularizing effects of intensified augmentation.

However, both models 2 and 3 overfit the limited dataset as evidenced by the gap between training and validation performance.

The validation performance indicated that the simpler baseline CNN (Model 1) generalized best on unseen data, achieving higher validation accuracy and AUC than the deeper models. The heavy augmentation in Model 2 reduced overfitting but slightly lowered discriminative performance, while Model 3's depth led to poor generalization and heavy overfitting due to insufficient data for effective learning.

Overall, these results demonstrate that for small medical imaging datasets, simpler architectures with minimal augmentation may outperform deeper networks with heavy augmentation.

Inference

The baseline CNN (Model 1) performed best on this dataset, highlighting the effect of dataset size on model complexity.

Regularization and augmentation, as shown in Model 2 and Model 3, still ended up causing overfitting as shown by the near-excellent performance during training and low performance during validation.

In conclusion, increasing model complexity without sufficient data can negatively impact generalization, highlighting the importance of balancing architecture depth, data preprocessing, and dataset size.

1.8 REFERENCES

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